COBOL Basics 2

Group Items/Records

```
WORKING-STORAGE SECTION.
01  StudentDetails    PIC X(26).
```

Group Items/Records

```
WORKING-STORAGE SECTION.
01  StudentDetails.
    02  StudentName       PIC X(10).
    02  StudentId      PIC 9(7).
    02  CourseCode     PIC X(4).
    02  Grant          PIC 9(4).
    02  Gender         PIC X.
```

Group Items/Records

```
WORKING-STORAGE SECTION.
01  StudentDetails.
    02  StudentName.
        03 Surname     PIC X(8).
        03 Initials       PIC XX.
    02  StudentId      PIC 9(7).
    02  CourseCode     PIC X(4).
    02  Grant          PIC 9(4).
    02  Gender         PIC X.
```

LEVEL Numbers express DATA hierarchy

- In COBOL, **level numbers** are used to decompose a structure into it's constituent parts.
- In this hierarchical structure the higher the level number, the lower the item is in the hierarchy. At the lowest level the data is completely atomic.
- The level numbers **01** through **49** are general level numbers but there are also special level numbers such as 66, 77 and **88**.
- In a hierarchical data description what is important is the **relationship** of the level numbers to one another, not the actual level numbers used.

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The level numbers 01 through 49 are general level numbers but there are also special level numbers such as 66, 77 and 88.

In a hierarchical data description what is important is the relationship of the level numbers to one another, not the actual level numbers used.

```
01 StudentDetails.
   02 StudentName.
      03 Surname      PIC X(8).
      03 Initials     PIC XX.
   02 StudentId       PIC 9(7).
   02 CourseCode      PIC X(4).
   02 Grant           PIC 9(4).
   02 Gender          PIC X.
```

```
01 StudentDetails.
   05 StudentName.
      10 Surname      PIC X(8).
      10 Initials     PIC XX.
   05 StudentId       PIC 9(7).
   05 CourseCode      PIC X(4).
   05 Grant           PIC 9(4).
   05 Gender          PIC X.
```

Group and elementary items.

- In COBOL the term "**group item**" is used to describe a data item which has been further subdivided.
  - A Group item is declared using
    cannot have a picture clause
  - Where a group item is the highe
    referred to as a **record** and u **01**
- The term "**elementary item**" is used to describe data items which are atomic; that is, not further subdivided.
- An elementary item declaration consists of;

must

- Every group or elementary item declaration **must** be followed by a full stop.

PICTUREs for Group Items

- Picture clauses are **NOT** specified for 'group' data items because the **size** a group item is the sum of the sizes of its subordinate, elementary items and its **type** is always assumed to be **PIC X**.
- The type of a group items is always assumed to be PIC X because group items may have several different data items and types subordinate to them.
- An X picture is the only one which could support such collections.

Assignment in COBOL

- In “strongly typed” languages like Modula-2, Pascal or ADA the assignment operation is simple because assignment is only allowed between data items with compatible types.
- The simplicity of assignment in these languages is achieved at the “cost” of having a large number of data types.
- In COBOL there are basically only three data types,

- But this simplicity is achieved only at the cost of having a very complex assignment statement.
- In COBOL assignment is achieved using the **MOVE** verb.

The MOVE Verb

$$\underline{\texttt{MOVE}} \begin{Bmatrix} \text{Identifier} \\ \text{Literal} \end{Bmatrix} \underline{\texttt{TO}} \ \{\text{Identifier}\} \ \ldots$$

- The **MOVE** copies data from the source identifier or literal to one or more destination identifiers.
- The source and destination identifiers can be group or elementary data items.
- When the destination item is alphanumeric or alphabetic (PIC X or A) data is copied into the destination area from **left** to **right** with space filling or truncation on the right.
- When data is MOVEd into an item the contents of the item are completely **replaced**. If the source data is too small to fill the destination item entirely the remaining area is **zero** or **space filled**.

MOVEing Data

```
MOVE "RYAN" TO Surname.
MOVE "FITZPATRICK" TO Surname.
```

C	O	U	G	H	L	A	N

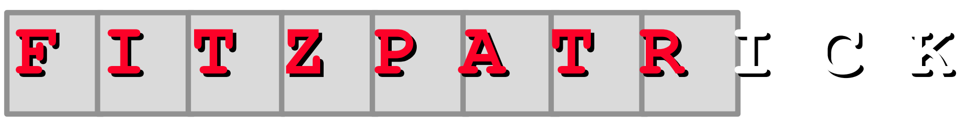
MOVEing Data

```
MOVE "RYAN" TO Surname.
MOVE "FITZPATRICK" TO Surname.
```

R	Y	A	N				

MOVEing Data

```
MOVE "RYAN" TO Surname.
MOVE "FITZPATRICK" TO Surname.
```



MOVEing to a numeric item.

- When the destination item is numeric, or edited numeric, then data is aligned along the **decimal point** with zero filling or truncation as necessary.
- When the decimal point is not explicitly specified in either the source or destination items, the item is treated as if it had an assumed decimal point immediately after its rightmost character.

```
01 GrossPay       PIC 9(4)V99.

MOVE ZEROS TO GrossPay.

MOVE 12.4 TO GrossPay.

MOVE 123.456 TO GrossPay.

MOVE 12345.757 TO GrossPay.
```

0 0 0 0 0 0

0 0 1 2 4 0

0 1 2 3 4 5 6

1 2 3 4 5 7 5 7

```
01 CountyPop        PIC 999.
01 Price         PIC 999V99.

MOVE 1234 TO CountyPop.

MOVE 12.4 TO CountyPop.

MOVE 154 TO Price.

MOVE 3552.75 TO Price.
```

1 2 3 4

0 1 2 4

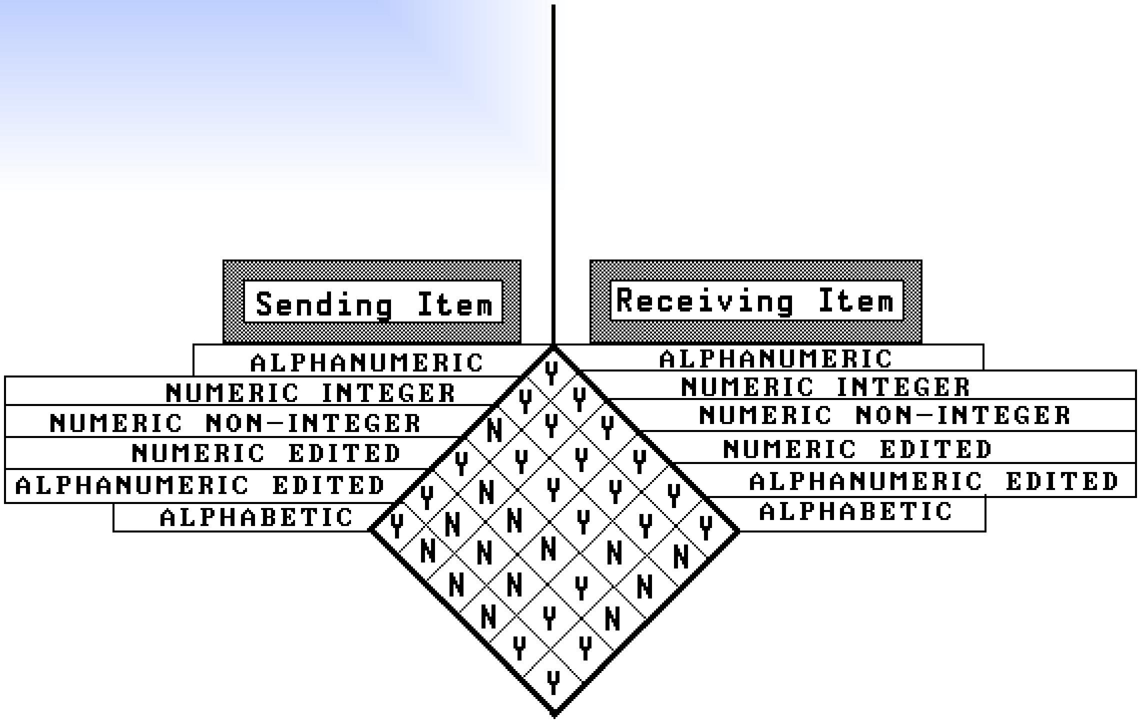
1 5 4 0 0

3 5 5 2 7 5

Legal MOVEs

Certain combinations of sending and receiving data types are not permitted (even by COBOL).

MOVE COMBINATIONS.



The DISPLAY Verb

$$\underline{\text{DISPLAY}} \begin{Bmatrix} \text{Identifier} \\ \text{Literal} \end{Bmatrix} \left[\begin{Bmatrix} \text{Identifier} \\ \text{Literal} \end{Bmatrix} \right] \ldots$$

$$\left[\underline{\text{UPON}} \text{ Mnemonic - Name} \right] \left[\text{WITH } \underline{\text{NO}} \ \underline{\text{ADVANCING}} \right]$$

- From time to time it may be useful to display messages and data values on the screen.
- A simple **DISPLAY** statement can be used to achieve this.
- A single DISPLAY can be used to display several data items or literals or any combination of these.
- The **WITH NO ADVANCING** clause suppresses the carriage return/line feed.

The ACCEPT verb

Format 1. <u>ACCEPT</u> Identifier [<u>FROM</u> Mnemonic - name]

$$\text{Format 2.}\quad \underline{\text{ACCEPT}}\ \text{Identifier}\ \underline{\text{FROM}} \begin{Bmatrix} \underline{\text{DATE}} \\ \underline{\text{DAY}} \\ \underline{\text{DAY-OF-WEEK}} \\ \underline{\text{TIME}} \end{Bmatrix}$$

```
01 CurrentDate        PIC 9(6).
* YYMMDD

01 DayOfYear          PIC 9(5).
* YYDDD

01 Day0fWeek          PIC 9.
* D (1=Monday)

01 CurrentTime        PIC 9(8).
* HHMMSSss    s = S/100
```

Run of Accept and Display program

```
Enter student details using template below
NNNNNNNNNSSSSSSSCCCCGGGGS
COUGHLANMS9476532LM511245M
Name is MS COUGHLAN
Date is 24 01 94
Today is day 024 of the year
The time is 22:23
```

```
      $ SET SOURCEFORMAT"FREE"
IDENTIFICATION DIVISION.
PROGRAM-ID.  AcceptAndDisplay.
AUTHOR.  Michael Coughlan.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 StudentDetails.
   02  StudentName.
       03 Surname       PIC X(8).
       03 Initials      PIC XX.
   02  StudentId        PIC 9(7).
   02  CourseCode       PIC X(4).
   02  Grant            PIC 9(4).
   02  Gender           PIC X.

01 CurrentDate.
   02  CurrentYear      PIC 99.
   02  CurrentMonth     PIC 99.
   02  CurrentDay       PIC 99.

01 DayOfYear.
   02  FILLER           PIC 99.
   02  YearDay          PIC 9(3).

01 CurrentTime.
   02  CurrentHour      PIC 99.
   02  CurrentMinute    PIC 99.
   02  FILLER           PIC 9(4).
```

```
PROCEDURE DIVISION.
Begin.
   DISPLAY "Enter student details using template below".
   DISPLAY "NNNNNNNNNSSSSSSSCCCCGGGGS                 ".
   ACCEPT  StudentDetails.
   ACCEPT  CurrentDate FROM DATE.
   ACCEPT  DayOfYear FROM DAY.
   ACCEPT  CurrentTime FROM TIME.
   DISPLAY "Name is ", Initials SPACE Surname.
   DISPLAY "Date is " CurrentDay SPACE CurrentMonth SPACE CurrentYear.
   DISPLAY "Today is day " YearDay " of the year".
   DISPLAY "The time is " CurrentHour ":" CurrentMinute.
   STOP RUN.
```